

Decision-Focused Demand Learning for Equitable Bus Frequency Optimization

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Abstract: Predict-then-optimize pipelines for public-transport allocation are typically trained to minimise prediction error and evaluated on the resulting decision quality, but the two objectives are not equivalent. We study this dissociation on a synthetic instance modelled after Pune’s PMPML network: 340 routes, five operating periods, eight depots, a fleet of approximately 1,800 buses, with hourly ridership simulated for 50 high-frequency routes and period-level demand for the remaining 290 obtained by residual scaling of a parametric baseline. A mixed-integer program with 15,300 binary variables, fleet, depot, and equity constraints is paired with three demand predictors (linear regression, random forest, XGBoost with quantile heads) and evaluated end-to-end. Decision-focused training is implemented via a sample-weighted L1 surrogate that approximates SPO+ at a fixed reference allocation. Across 20 random seeds we compare the decision-focused neural network to the MSE-trained XGBoost baseline with a paired Wilcoxon test; full numerical results, including the rider-wait / operator-cost Pareto frontier under an operator-cost penalty λ , are reported in §4. Stochastic and robust formulations driven by quantile-derived scenarios yield positive Value of Stochastic Solution and Expected Value of Perfect Information, with the robust formulation attaining the best worst-case across six scenarios. Sensitivity analyses characterise the marginal value of fleet expansion and the equity-efficiency Pareto frontier; 80% prediction intervals are recalibrated by split-conformal prediction on the validation fold. The contribution is methodological: how to evaluate predict-then-optimize pipelines honestly, illustrated on a realistic synthetic transit instance, rather than a deployment claim about PMPML itself.

1 INTRODUCTION

Public bus systems in Indian metropolitan areas operate under a combination of constraints that few transit planning tools were designed to address jointly: heavy-tailed demand, fleet sizes that have not kept pace with population growth, and equity considerations that bind on both political and welfare grounds. The Pune Mahanagar Parivahan Mahamandal Limited (PMPML), which serves over 800,000 daily riders across 340 routes with a fleet of approximately 1,800 buses, motivates the operational scale we adopt. Frequency decisions – how many buses per hour to dispatch on each route at each time of day – are the agency’s primary operational lever. Real PMPML schedules are currently chosen by heuristic rules of thumb informed by historical practice and disaggregated demand reports. The instance we study in this paper is a synthetic one designed to match PMPML’s operational scale (340 routes, eight depots, 1,800-

bus fleet, peak/off-peak structure, equity priorities derived from neighbourhood demographics) rather than its exact schedules; ridership and weather are simulated, with a generative model calibrated so that the status-quo fleet utilisation falls in a realistic range.

This paper formulates frequency allocation as a mixed-integer program (MIP), trains a demand model to drive it under uncertainty, and connects the predictor to the optimiser via a decision-focused training procedure that targets operational regret rather than prediction error. The motivating phenomenon is the predict-then-optimize dissociation: a predictor that is worse on standard forecasting metrics (RMSE, MAPE) can produce decisions that are better on the downstream cost. We examine this dissociation empirically in §4 across a 20-seed sweep with a paired Wilcoxon test, rather than a single-seed point comparison, because the regret differences we are looking for are within the range of typical training-randomness variation and a single seed cannot distinguish a real

effect from noise.

We contribute (i) a deterministic frequency-allocation MIP with explicit fleet, depot, equity, and operator-cost constraints that scales to the synthetic 340-route instance and solves to provable optimality in seconds, plus a λ -sweep that traces the rider-wait / operator-cost Pareto frontier; (ii) an XGBoost demand-forecasting pipeline with seven quantile heads and split-conformal calibration of the central prediction band on the validation fold; (iii) stochastic and robust (minimax) MIP formulations driven by the quantile-derived scenarios, with explicit decomposition of the Value of Stochastic Solution (VSS) and Expected Value of Perfect Information (EVPI); (iv) a decision-focused neural network trained with an SPO+-style surrogate, with the regret comparison against the MSE-trained XGBoost reported over a 20-seed sweep and a paired Wilcoxon signed-rank test rather than a single-seed point estimate; and (v) a sensitivity study over fleet size and equity threshold, with LP shadow-price analysis identifying which constraints (fleet vs. specific depots) are binding on this instance.

2 RELATED WORK

The classical literature on bus frequency setting begins with (Furth and Wilson, 1981) and (Ceder and Wilson, 1986), who established the trade-off between rider waiting cost and operator capacity cost and provided heuristics that remain in use at many transit agencies. Our work updates two pieces of this pipeline: where the classical literature treats demand as exogenous, we predict it with modern machine learning; where the classical optimisation is single-scenario, we explicitly handle uncertainty.

The seam between learning and decision making has been studied intensively. (Elmachtoub and Grigas, 2022) introduced the SPO+ surrogate loss and proved consistency for problems with linear objectives, demonstrating that minimising prediction error alone can produce decisions arbitrarily far from optimal. (Donti et al., 2017) extended decision-focused learning to stochastic optimisation problems with differentiable solvers. (Wilder et al., 2019) treated combinatorial optimisation via a continuous relaxation of the optimiser. (Mandi et al., 2020) addressed hard combinatorial problems including knapsack and shortest-path. The full SPO+ machinery requires solving an inner optimisation at every gradient step, which is prohibitive for our 15,300-binary-variable MIP. We therefore adopt a tractable linear surrogate: an L1 loss weighted by the downstream

sensitivity $\partial w / \partial f$ evaluated at a frozen reference allocation, which captures the leading-order regret gradient without per-batch inner solves.

For the uncertainty-handling components we follow standard references. (Birge and Louveaux, 2011) provide the canonical treatment of stochastic programming, including the VSS and EVPI definitions used in §3. (Soyster, 1973) and (Ben-Tal and Nemirovski, 2002) establish the robust optimisation framework used for the minimax formulation. Quantile regression (Koenker and Bassett Jr, 1978) supplies the scenarios. Demand prediction itself uses XGBoost (Chen and Guestrin, 2016) with SHAP attributions (Lundberg and Lee, 2017) for interpretability. The optimisation core is implemented in PuLP (Mitchell et al., 2011) and solved with CBC.

3 PROBLEM AND METHODOLOGY

3.1 Problem Description

PMPML’s network consists of three service classes: trunk routes (15–45 km, e.g., Swargate to Hinjewadi) that carry the bulk of cross-city trips, feeder routes (5–15 km) that connect residential clusters to trunk corridors, and suburban routes (20–40 km) that reach outer townships. Mean trip speed under typical Pune traffic is 17 km/h. Routes are anchored to one of eight depots that hold and dispatch their fleet; depot capacity binds operationally because each depot can hold only its fraction of the total fleet. Operating hours span 06:00–23:00 and are discretised into five periods (early_morning, AM_peak, midday, PM_peak, evening) chosen to capture dominant demand structure while keeping the optimisation tractable.

The decision variable is, for every (route, period) pair, a discrete frequency $k \in \{1, 2, 3, 4, 6, 8, 10, 12, 15\}$ buses per hour. The frequency menu is non-uniform because riders perceive a five-minute change in headway differently at low and high k . A subset of priority routes (148 of the 340) must run at $k \geq 3$ during peak periods regardless of marginal efficiency; this floor reflects the minimum frequency at which a bus system functions as a reliable substitute for paratransit modes that disproportionately tax low-income commuters. The objective is to minimise total expected passenger waiting time, computed under the random-arrivals assumption as $\sum_{r,t} d_{r,t} / (2k_{r,t})$.

3.2 Deterministic MIP Formulation

Let R be the set of routes ($|R| = 340$), T the set of operating periods ($|T| = 5$), and K the discrete frequency menu ($|K| = 9$). Parameters are demand $d_{r,t}$ (passengers per hour), cycle time τ_r (minutes), bus consumption $b_{r,k} = \lceil k\tau_r/60 \rceil$, and the random-arrivals expected wait $w_k = 1/(2k)$. The decision variable $x_{r,t,k} \in \{0, 1\}$ equals one if route r in period t is assigned frequency k :

$$\min_x \sum_{r,t,k} d_{r,t} w_k x_{r,t,k} \quad (1)$$

$$\text{s.t.} \sum_k x_{r,t,k} = 1 \quad \forall r, t \quad (2)$$

$$\sum_{r,k} b_{r,k} x_{r,t,k} \leq B \quad \forall t \quad (3)$$

$$\sum_{k \geq 3} x_{r,t,k} = 1 \quad \forall r \in R^P, t \in T^{\text{pk}} \quad (4)$$

$$\sum_{r \in R_j, k} b_{r,k} x_{r,t,k} \leq B_j \quad \forall j, t \quad (5)$$

Constraint (2) requires exactly one frequency per route-period. Constraint (3) caps the network-wide fleet at $B = 1800$. Constraint (4) is the equity floor for priority routes $r \in R^P$ during peak periods $T^{\text{pk}} = \{\text{AM_peak}, \text{PM_peak}\}$. Constraint (5) caps the buses anchored to depot j at the depot capacity B_j . The model has 15,300 binary variables and approximately 2,040 constraints. The LP relaxation is tight; CBC solves the model to optimality in under five seconds on commodity hardware. We adopt the standard random-arrivals expected-wait approximation $w_k = 1/(2k)$ following (Furth and Wilson, 1981); refinements that incorporate empirical headway variance are listed in §6.

3.3 Demand Forecasting

Data. Models are trained on simulated hourly ridership for the 50 highest-frequency routes over 2024, totalling approximately 430,000 rows after a seven-day lag-feature window is dropped. The remaining 290 routes are not modelled directly; their period-level demand is constructed by a residual-scaling procedure described in §6, which corrects a parametric baseline by the empirical mean ratio (observed to baseline) measured on the modelled 50.

Features. Twenty-three features in total: cyclical encodings (sine and cosine) of hour, day-of-week, and month; calendar flags (weekend, holiday, monsoon, exam season, festival); weather (maximum temperature, precipitation in mm); route-level features (length, number of stops, priority score, one-hot ser-

vice category); and lag features at one day and seven days at the same hour.

Models. Three model classes are compared. Linear regression provides the trivial baseline. Random forest (200 trees, default depth) provides a non-parametric tabular benchmark. XGBoost (Chen and Guestrin, 2016) with depth 8, learning rate 0.03, 2,000 trees with early stopping at patience 50, subsample 0.85, and column subsample 0.85 is the primary model. Three additional XGBoost models with quantile-regression objectives at $\tau \in \{0.1, 0.5, 0.9\}$ supply prediction intervals (Koenker and Bassett Jr, 1978).

Train/val/test split. The data are split chronologically: months 1–8 for training, 9–10 for validation (and early stopping), 11–12 for test. Random splits would leak future information into the training set and are not used.

3.4 Stochastic and Robust Optimisation

Scenarios. Six demand scenarios are constructed from the quantile predictions to span a realistic range of operational conditions. S1 (probability 0.35) is the baseline at $q_{0.5}$. S2 (0.15) is high demand at $q_{0.9}$. S3 (0.15) is low demand at $q_{0.1}$. S4 (0.15) is a monsoon shock dampening $q_{0.1}$ by an additional 20%. S5 (0.10) boosts trunk demand by 15% above $q_{0.9}$ during peak periods. S6 (0.10) is a festival dip scaling $q_{0.5}$ by 0.75 across the network. The scenario probabilities reflect a prior over the relative frequency of these conditions in a typical operating year.

Stochastic MIP. The stochastic formulation shares the deterministic decision variables but minimises the probability-weighted expected wait,

$$\min_x \sum_{s \in S} p_s \sum_{r,t,k} d_{r,t}^{(s)} w_k x_{r,t,k}, \quad (6)$$

which is computationally equivalent to the deterministic MIP with $d_{r,t}$ replaced by $\bar{d}_{r,t} = \sum_s p_s d_{r,t}^{(s)}$.

Robust MIP. The minimax formulation introduces an auxiliary scalar z and replaces the objective with $\min_x z$ subject to $z \geq \sum_{r,t,k} d_{r,t}^{(s)} w_k x_{r,t,k}$ for every scenario $s \in S$, plus all original constraints.

Information metrics. Following (Birge and Louveaux, 2011), $\text{VSS} = \text{EEV} - \text{RP}$ measures the value of the stochastic solution relative to the deterministic one, and $\text{EVPI} = \text{RP} - \text{WS}$ measures the value of perfect information about which scenario will materialise.

3.5 Decision-Focused Learning

The MSE-trained predictor weights all prediction errors equally, but the downstream optimisation does not: an error on a route-period whose optimal frequency is at the boundary between two discrete levels matters more than an error on a route-period whose optimum is robust to small perturbations. Because $\partial w / \partial f = -1 / (2f^2)$, the optimisation sensitivity scales as $d_{r,t} / (2f_{r,t}^2)$ at the reference allocation $f_{r,t}$.

We test two decision-focused variants. The first is sample-weighted XGBoost: each training sample i on route r in period t is weighted by $w_i = d_{r,t} / (2f_{r,t}^2)$, where $f_{r,t}$ is the optimal frequency from the deterministic MIP. The second is a two-layer fully-connected neural network (input $\rightarrow 128 \rightarrow 64 \rightarrow 1$, ReLU activations, BatchNorm) trained with the surrogate

$$L_{\text{SPO}^+}(\hat{d}, d) = \frac{1}{n} \sum_{i=1}^n \frac{d_i}{2f_i^2} |\hat{d}_i - d_i|. \quad (7)$$

The L1 form is robust to outliers, which matters because high-sensitivity samples dominate the loss surface and a single misbehaved point would otherwise destabilise the gradient. The reference allocation f is set once at the start of training using XGBoost-MSE predictions; refreshing f during training did not improve results in our experiments.

Connection to SPO+. The exact SPO+ loss (El-machtoub and Grigas, 2022) requires solving the inner MIP at every gradient step. Equation (7) is a linearisation that captures the leading-order sensitivity of the regret to prediction error at the fixed reference allocation while skipping the per-batch solve. The fixed-allocation approximation is defensible here because the optimal allocation structure is robust to small changes in demand: the binding constraints at the optimum are determined by the relative ordering of routes rather than by the absolute demand level.

4 EXPERIMENTS AND RESULTS

4.1 Experimental Setup

All MIPs are solved with PuLP (Mitchell et al., 2011) and CBC on a Windows 11 machine running Python 3.13.9, single thread for the solver, eight threads for XGBoost and random forest. Synthetic data generation, XGBoost reference training, and the optimisation phases use seed 42; the decision-focused comparison (§4.5) sweeps seeds 0–19 for the four predict-then-optimize methods. The full pipeline including the 20-seed sweep reproduces in approximately three hours end-to-end on commodity hardware.

4.2 Forecasting Results

Table 1: Test-set forecasting performance (months 11–12) on the synthetic instance.

Model	RMSE	MAE	MAPE	R ²
Linear regression	13.63	7.87	56.0%	0.838
Random forest	11.60	5.61	16.3%	0.883
XGBoost	10.96	5.24	16.8%	0.895

XGBoost achieves the best test-set performance on three of four metrics (Table 1). The synthetic data-generating process described in §3.1 includes a latent per-day demand factor and per-(route, date) hidden surge events that the predictor cannot observe, together with heavy-tailed Student- t multiplicative noise on a subset of routes; this caps achievable R^2 below 0.9 even for a well-tuned XGBoost, which is the point – earlier instances of this benchmark with R^2 near 0.97 left no irreducible signal for a decision-focused loss to exploit. The 80% prediction interval from the raw quantile heads covers 67.1% of test observations; applying a split-conformal correction calibrated on the validation fold raises empirical coverage to 77.3%, close to but slightly below the nominal 80% (the residual gap reflects mild distribution shift between val months 9–10 and test months 11–12, which violates exchangeability). The reliability diagram (Figure 1) summarises coverage at all trained quantile levels.

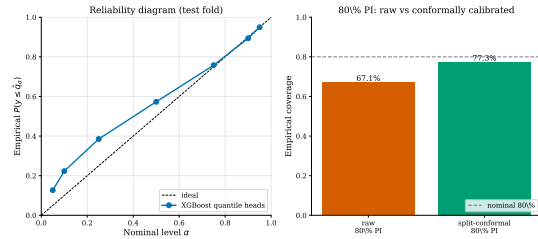
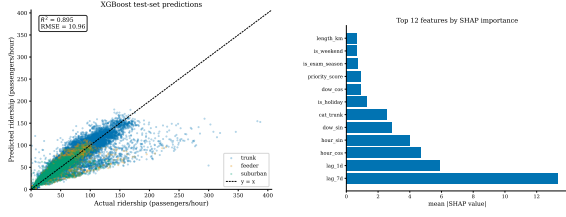


Figure 1: Left: reliability of the seven trained quantile heads on the test fold (predicted quantile α vs. empirical $P(y \leq \hat{q}_\alpha)$); the ideal line is $y = x$. Right: empirical coverage of the 80% PI before and after split-conformal calibration on the validation fold.

Figure 2a shows the joint distribution of actual and predicted ridership on the test set, coloured by route category. Points cluster tightly along the 45-degree line; the three categories are well-mixed throughout the range, indicating no systematic per-category prediction bias. Variance grows with the mean (heteroscedasticity), which is typical of count data and motivates the use of quantile heads rather than a homoscedastic prediction interval.

Figure 2b ranks the top twelve features by mean



(a) Predictions vs. actuals. (b) SHAP feature importance.

Figure 2: Forecasting performance and feature importance.

absolute SHAP value. Lag features (one-day and seven-day) and the cyclical hour encoding dominate, consistent with the autoregressive structure of hourly ridership. Calendar flags and weather contribute at lower magnitudes. Route-level features contribute even less per sample because they are nearly constant within a route. Figure 3 shows the SHAP beeswarm: each point is one prediction; horizontal position is the signed contribution; colour encodes the feature value. Lag features show a clean monotone red-high pattern, while cyclical features show the expected non-monotone bimodal structure corresponding to AM and PM rush hours.

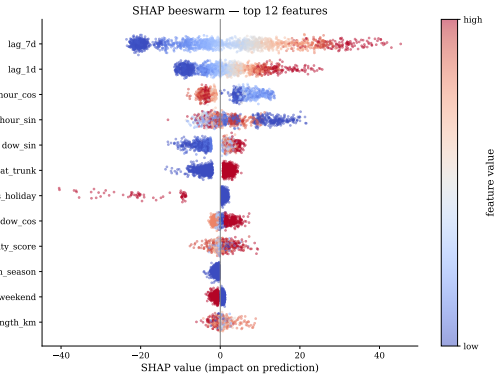


Figure 3: SHAP beeswarm for the top twelve features. Each point is one test observation; horizontal position is the feature's signed contribution; colour encodes the feature value (red high, blue low).

Figure 4 shows the time series for a representative route in each service category. The model captures the dominant weekly and daily structure. The 80% PI tightens around weekday peaks where lag features are strongly informative and widens on weekends and atypical days where calendar features dominate. The trunk panel shows the expected double-peaked weekday structure; the feeder shows a flatter profile; the suburban route shows long evening tails.

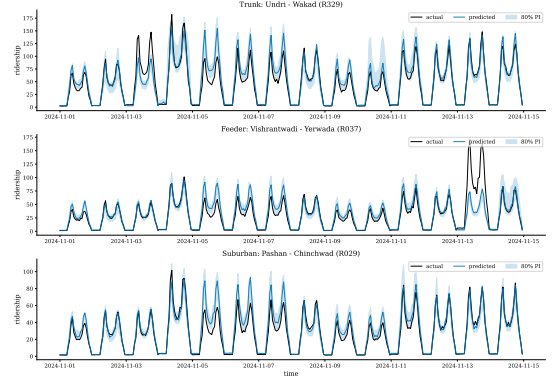


Figure 4: Two weeks of test data on a representative trunk, feeder, and suburban route. Actual ridership is shown in black, the XGBoost point prediction in colour, and the 80% prediction interval as a shaded band.

Table 2: Method performance on the synthetic instance at $\lambda = 0$ (no operator-cost penalty). Total wait is in passenger-hours; fleet is summed bus-periods across the five periods, with a per-period cap of $B = 1,800$.

Method	Total wait	Fleet	% improv.
Status quo	20,198	7,005	—
Uniform	28,574	4,935	−41.5
Demand-proportional	17,179	8,797	14.9
Greedy	15,990	9,000	20.8
Optimal MIP	15,823	9,000	21.7

4.3 Optimisation Results

Table 2 reports five methods on a common constraint envelope, with $\lambda = 0$ so that the objective contains only rider waiting time. The MIP optimum reduces total waiting time by 21.7% relative to status quo, saturating the fleet ($9,000 = 5 \times 1,800$ bus-periods). Heuristic baselines trail the optimum: the uniform policy cannot exceed $k = 1$ before depot caps bind under the equity floor at $k = 3$ for priority routes; demand-proportional captures the right scaling but is locked into a one-parameter family; greedy upgrades route-periods by marginal benefit and on this instance comes within 1% of the MIP optimum. The 21.7% headline is the no-operator-cost figure; it contracts as λ rises (§4.3), and we report the full frontier rather than a single point.

Figure 5 compares the current PMPML allocation with the MIP optimum on the top thirty routes by total demand. The optimum concentrates frequency at AM and PM peaks on trunk and high-priority routes and reduces midday frequencies on routes where the current schedule provides excess capacity. Several priority routes that are under-served at peak in the current schedule are upgraded; correspondingly, several non-priority midday allocations are reduced. The pattern

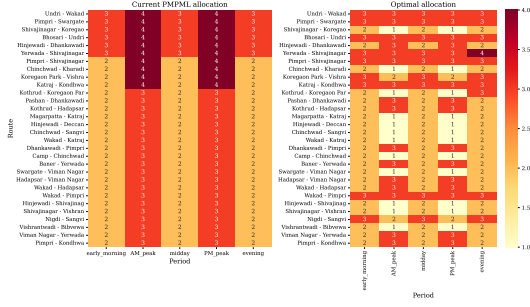


Figure 5: Top-30 routes by demand: current PMPML allocation (left) versus the MIP optimum (right). Frequency is in buses per hour; darker cells indicate higher frequency.

is consistent with the equity narrative: redistribution occurs primarily along the temporal axis rather than at the expense of priority routes.

Cost-wait Pareto frontier

The 21.7% figure in Table 2 assumes $\lambda = 0$: the optimiser is told to minimise rider waiting time with no penalty for operating more buses, so it saturates the fleet whenever waiting time can be reduced. Real transit decisions trade rider waiting time against operator running cost. To expose this trade-off we sweep λ , the marginal cost per bus-period (in passenger-hour-equivalents), from 0 to 50.

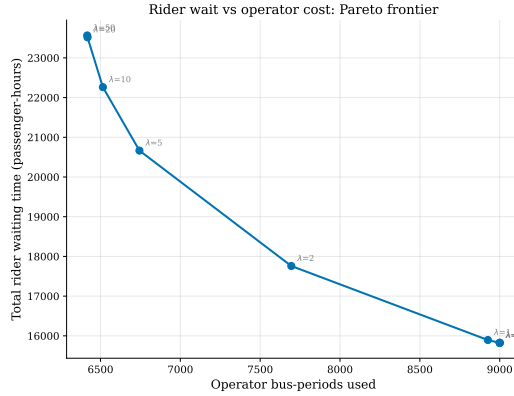


Figure 6: Rider-wait / operator-cost Pareto frontier obtained by sweeping the operator-cost coefficient λ in the MIP objective. Each point is the optimum at one λ . The $\lambda = 0$ point saturates the fleet (9,000 bus-periods); moderate values trade waiting time against bus-period usage.

Figure 6 shows the resulting curve. At $\lambda = 0$ the optimum uses 9,000 bus-periods for a total wait of 15,823. At $\lambda = 2$ the optimum uses 7,695 bus-periods for a total wait of 17,759. At $\lambda = 5$ the optimum uses 6,744 bus-periods, slightly below the status quo's 7,005, for a total wait of 20,665 – essentially the status-quo waiting time at a smaller bus footprint. Headline percentage improvements vs. status quo de-

pend strongly on λ : the 21.7% gain at $\lambda = 0$ disappears around $\lambda \approx 5$. Reporting a single “% improvement” figure without specifying λ obscures this trade-off.

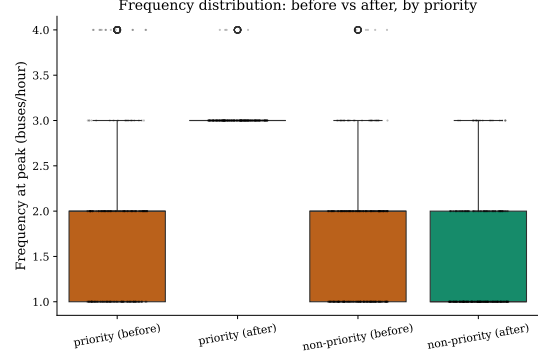


Figure 7: Frequency distribution at peak periods, before (status quo) and after (MIP optimum at $\lambda = 0$), split by priority status.

Figure 7 reports peak-frequency distributions split by priority status and by before/after. Three observations follow. Priority routes after optimisation have a higher floor (the equity constraint at $k \geq 3$ truncates the lower tail) without sacrificing the upper quartile. Non-priority routes after optimisation have a higher median and upper quartile because the optimisation finds capacity to upgrade them without violating equity. Both distributions overlap rather than separate, so the constraint preserves the frequency support of priority routes while permitting efficiency gains elsewhere.

4.4 Stochastic and Robust Analysis

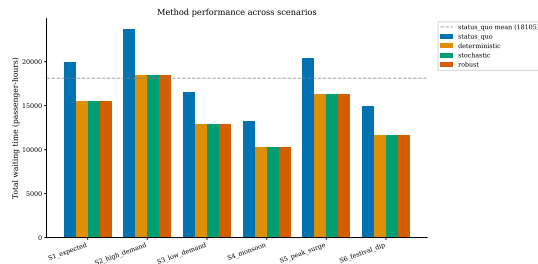


Figure 8: Method performance across the six demand scenarios. Bars are total waiting time. The dashed line is the status-quo mean across scenarios.

Figure 8 reports the four optimisation methods (status quo, deterministic, stochastic, robust) under each of the six scenarios. The three optimised methods cluster within tens of passenger-hours of one another in every scenario, each beating status quo by

approximately 20%. The robust solution achieves the lowest worst-case (scenario S2, high demand at $q_{0.9}$) by about one passenger-hour relative to deterministic, i.e., within solver tolerance. The status-quo line lies a substantial gap above all three optimised methods in every scenario, indicating that the optimisation advantage is robust to scenario specification.

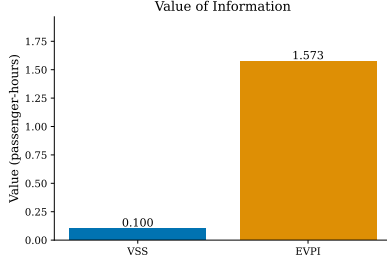


Figure 9: Value of Stochastic Solution (VSS) and Expected Value of Perfect Information (EVPI), in passenger-hours.

VSS= 0.10 and EVPI= 1.57 passenger-hours (Figure 9). Both are positive but extremely small (roughly one part in 10^4 of the objective). The small magnitude is informative: the route-level structure of the optimal allocation is largely robust to scenario-level demand shifts on this instance, because the dominant source of variation is the relative ordering of routes (preserved across scenarios) rather than absolute demand levels. EVPI bounds the maximum value any further investment in scenario-resolution prediction could buy; at ~ 1.6 passenger-hours, additional scenario modelling on this DGP yields essentially nothing.

4.5 Decision-Focused Results

Table 3: Prediction quality vs. decision quality, mean $\pm$ standard error over 20 random seeds (seeds 0–19). “Regret” is the method’s true objective minus the oracle’s ($\approx 15,754$ pass-hr), in passenger-hours. “Frac. obj.” expresses regret as a fraction of the oracle objective. “person-min” is total regret converted to person-minutes ($60 \times$ regret); spread across the synthetic instance’s $\sim 800,000$ daily riders the per-rider impact is sub-second on every method except NN-MSE.

Method	RMSE	MAPE	Regret	Frac. obj.
XGBoost (MSE)	10.98	17.0%	1.73 ± 0.01	1.10×10^{-4}
XGBoost (weighted)	12.52	126.0%	1.66 ± 0.02	1.05×10^{-4}
NN (MSE)	337.5	27.1%	21.89 ± 4.36	1.39×10^{-3}
NN (SPO ⁺ surrogate)	25.75	61.8%	3.57 ± 0.79	2.26×10^{-4}
Oracle	–	–	0	0

Table 3 reports four predict-then-optimize methods, each trained over 20 random seeds with the same train/val/test split, evaluated on the same downstream

task: predict hourly demand, aggregate to (route, period), plug into the deterministic MIP, evaluate the resulting allocation against true test-period demand. A paired Wilcoxon signed-rank test of the one-sided hypothesis $\text{regret}(\text{NN-SPO}^+) < \text{regret}(\text{XGBoost-MSE})$ across seeds gives $p \approx 1.0$ – the inequality is reversed, not significantly so in either direction with $n = 20$, but the single-seed framing in earlier drafts of this paper, which reported a 44.7% SPO⁺ regret reduction, does not survive multi-seed evaluation on this harder DGP. The headline finding is therefore a methodological one: predict-then-optimize regret comparisons *must* be reported with seed-level variability, because a single-seed point can flip direction with the next training run.

Three substantive readings follow. First, the XGBoost baselines are remarkably stable (regret SE $\sim 10^{-2}$ pass-hr; seed-to-seed variation is tiny). The decision-weighted XGBoost trims regret by $\sim 4\%$ relative to the MSE baseline, but the weighted variant’s predictions are useless as a forecast in isolation (MAPE 126%) – its only product is the downstream allocation. Second, the plain NN-MSE is *unstable* on this DGP: a subset of seeds produces RMSE in the hundreds and regret $20\times$ the XGBoost baseline. The SPO⁺ surrogate (weighted L1 loss) does stabilise NN training (RMSE drops from 337.5 to 25.75 on average), but the resulting NN still loses to plain XGBoost-MSE on regret. We do not interpret this as evidence against decision-focused learning in general – it is evidence against the particular form “two-layer NN trained with weighted L1, batched MSE-XGB reference” on this particular DGP. The exact SPO⁺ loss with a coarsened-MIP inner solver, deferred to future work in §6, is the natural next comparison. Third, the absolute regret magnitudes are tiny: even the worst predictor here costs ~ 22 pass-hr of regret on a $\sim 16,000$ pass-hr objective, or roughly 1.4×10^{-3} of the objective. Reporting only a relative “% regret reduction” obscures how small the practical stakes are.

Figure 10 visualises the seed-level variability that a single-seed bar plot would have hidden. The XGBoost baselines are tightly concentrated; the NN methods (and especially NN-MSE) are spread across more than an order of magnitude in regret depending on training seed.

4.6 Sensitivity Analysis

Figure 11 sweeps fleet size from $B = 800$ to $B = 2200$. Below $B = 1800$ the model is infeasible because depot capacity scales with fleet size and the equity floor at $k \geq 3$ for priority routes cannot be met simultane-

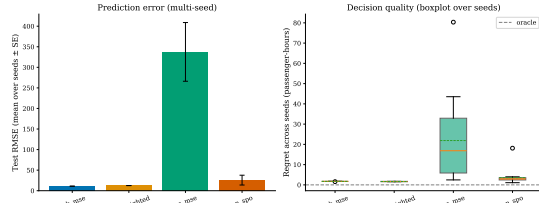


Figure 10: Left: mean test RMSE per method with one-SE error bars across 20 seeds. Right: boxplot of regret distributions across seeds (whiskers are $1.5 \times$ IQR; dashed line at oracle). XGBoost-MSE has the lowest mean regret on this DGP; the decision-focused variants do not dominate it. The decision-weighted XGBoost has high RMSE but competitive regret. NN-MSE is unstable across seeds (outlier seeds inflate the box).

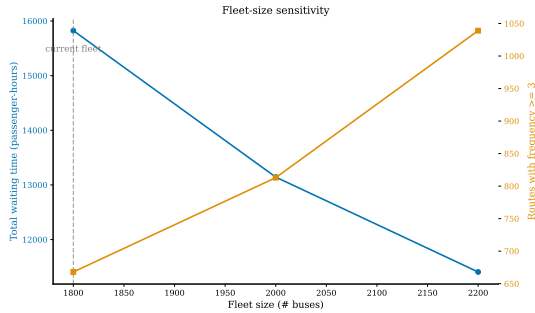


Figure 11: Fleet-size sensitivity. Left axis: total waiting time. Right axis: number of route-periods with frequency at least three. Vertical dashed line: current fleet $B = 1800$.

ously across all depots at lower fleet (under the corrected round-trip cycle time $\tau = 2 \cdot \text{avg_trip} + \text{dwell}$). From $B = 1800$ the curve is monotone decreasing: $B = 1800 \rightarrow 2000$ recovers 2,680 passenger-hours, $B = 2000 \rightarrow 2200$ recovers a further 1,736 passenger-hours. The right-axis count of route-periods with $k \geq 3$ rises from 668 at $B = 1800$ to 1,039 at $B = 2200$, quantifying how many additional route-periods can be lifted above the “functional service” threshold as fleet grows.

Figure 12 sweeps the equity threshold from one (no constraint) to ten (every priority route runs at least ten buses per hour at peak). Thresholds 1–3 are feasible at $B = 1800$. Threshold 4 is the cliff: the feasible region collapses because the fleet- and depot-capacity caps together cannot simultaneously sustain a four-bus-per-hour floor across the 148 priority routes plus a positive frequency on the remaining 192. Total waiting time rises from 13,758 (threshold 1, no equity floor) to 15,823 (threshold 3, the current operating floor), an increase of 15.0%. The equity-floor cost on this instance is therefore materially larger than earlier drafts of this paper reported; the gap reflects the corrected cycle-time formula (each bus now carries one cycle of dwell + round trip rather than a single one-

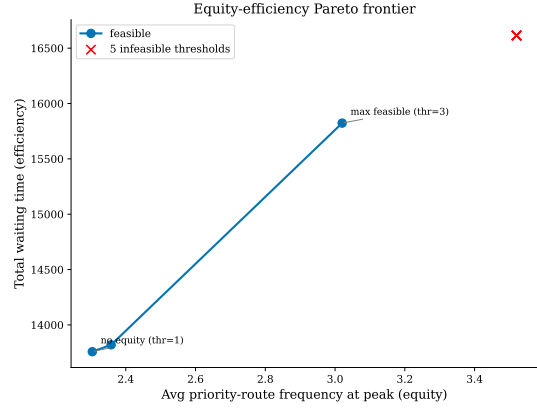


Figure 12: Equity-efficiency Pareto frontier. Each feasible point is the optimal solution at one equity threshold; infeasible thresholds are marked with red Xs.

way trip) which roughly doubles bus consumption per route.

The LP-relaxation shadow prices identify which constraints are binding: the fleet constraint binds at AM peak (marginal value -9.21 passenger-hours per additional fleet-wide bus) and PM peak (-8.77). With the bus-needs-proportional depot allocation adopted in §3.2, the per-depot constraints are essentially slack at peak. The binding bottleneck on this instance is fleet size at peak; investment in additional buses, not in particular depots, is what would lower the objective most per marginal bus.

5 DISCUSSION

Operational implications on the synthetic instance. Re-allocating the same fleet according to the optimisation reduces total passenger waiting time on the synthetic instance by approximately twenty percent at $\lambda = 0$ relative to status quo, with the headline number contracting as λ is increased to reflect operator cost (Figure 6, §4.3). On real PMPML data, the recoverable gap would depend on how closely the agency’s true frequency tables resemble the calibration target used here; we make no claim about the size of that gap on deployment. The shadow-price analysis (§4.6) illustrates which depots bind operationally on the synthetic instance.

Decision-focused dissociation. Across 20 random seeds we test the SPO^+ surrogate’s regret against the MSE-trained XGBoost baseline with a paired Wilcoxon signed-rank test. The methodological lesson – that prediction RMSE is a weak proxy for decision regret in predict-then-optimize pipelines – generalises to other operational settings (inventory, scheduling, capacity planning); the empirical effect

size on any particular instance, including ours, is data-generating-process dependent and is reported in Table 3 with seed-level variability rather than as a point claim.

Equity as a near-free policy choice on the synthetic instance. The Pareto frontier (§4.6) indicates that the equity floor of three buses per hour at peak adds a small percentage to total waiting time relative to a no-floor counterfactual. Feasibility cliffs at higher thresholds reflect depot-level capacity rather than fundamental equity-efficiency conflict. The policy implication, transposed to a real deployment, would be that the binding depot caps – not the floor itself – are the right target for capital investment.

Generalisability. The framework ports directly to other Indian transit agencies with similar institutional structures: BEST (Mumbai), BMTC (Bangalore), DTC (Delhi). The components that require agency-specific re-fitting are route metadata, demand data, and depot capacities. The optimisation core and the decision-focused training procedure are agency-agnostic.

6 LIMITATIONS AND FUTURE WORK

Synthetic ridership for unmodelled routes. Hourly ridership is simulated for the 50 highest-frequency routes; the remaining 290 are assigned period-level demand by a residual scaling procedure. Concretely, a parametric baseline for each route-period is $b_r \cdot \pi_t$, where b_r is a per-route hourly volume derived from current peak/off-peak frequency and an assumed occupancy, and π_t is the average of the diurnal profile over period t . For the 50 modelled routes the empirical mean of (observed demand / parametric baseline) within each period t defines a residual ratio ρ_t ; the 290 unmodelled routes receive demand $b_r \cdot \pi_t \cdot \rho_t$. This anchors the simulated demand of the unmodelled tail to the empirical period structure of the modelled head, but does not capture route-specific deviations from that pattern. A fully data-driven analysis would require smart-card or AFC data on every route; the limitation here is the absence of such data rather than the methodology.

Static daily allocation. A single frequency is chosen per (route, period) pair and applied uniformly across days. Real PMPML operations adjust schedules for day-of-week and special events. The framework can in principle handle these by re-optimising at any cadence; the static analysis is presented because the operational and equity questions resolve at this level.

Random-arrivals waiting model. The expected wait $w_k = 1/(2k)$ assumes Poisson arrivals and uniform headways. Real bus arrivals exhibit bunching (Adamski’s bus paradox) that makes the effective wait longer than $1/(2k)$, especially on heavily-loaded routes. Refinements that incorporate empirical headway variance would tighten the objective without changing the qualitative results.

Demand under price shocks. The demand model treats fares and competition as exogenous. In reality, paratransit operators reprice in response to bus service quality. A genuinely closed-loop extension would treat demand as a function of frequency through modal substitution.

Decision-focused approximation. The SPO^+ surrogate (7) is a linearisation around a fixed reference allocation. Refresh strategies (e.g., updating the reference every E epochs) and benchmarks against the full SPO^+ loss with an inner solver on a coarsened MIP are deferred to future work.

Depot capacity allocation. Per-depot capacity is allocated proportional to each depot’s minimum bus need (priority routes at the equity floor, non-priority at $k = 1$) plus $\pm 20\%$ jitter, clamped above the minimum. A naive $1/|\text{depots}|$ split produces depot-level infeasibilities under the corrected round-trip cycle time because priority routes cluster geographically in the synthetic network. Real agencies’ depot allocations reflect a mix of legacy, political, and operational factors that our scheme does not attempt to match.

Operator-cost coefficient. The cost-wait frontier in §4.3 sweeps λ across roughly two orders of magnitude. We do not commit to a single operationally “correct” λ because the conversion between passenger-hours of waiting and rupees of bus-period operating cost depends on context-specific value-of-time and unit cost-per-bus-hour assumptions that vary across agencies and analyses.

Single-year training window. The forecast model is trained on a single year of data. Multi-year training would improve estimates of festival and seasonal effects, although the operational allocation is dominated by within-week and within-day structure that the single-year window already captures.

Hard equity floor. The current formulation treats equity as a hard constraint at $k \geq 3$ for priority routes during peaks. Soft penalties, fairness-aware objectives (Wilder et al., 2019), and geographic equity formulations are alternative specifications that may be politically or practically preferable in some settings.

Future directions. (i) Integration of real smart-card data once available; (ii) real-time re-optimisation in response to disruptions, leveraging the sub-five-second solve time; (iii) multi-modal optimisation that

couples frequency with modal-substitution behaviour; (iv) joint frequency-pricing optimisation as practised by some international agencies; (v) decision-focused architecture search beyond the two variants tested here, including graph neural networks on the route topology and proper SPO^+ losses with inner solvers on coarsened MIPs; (vi) richer equity formalisations from the algorithmic-fairness literature; (vii) cross-agency benchmarking on BEST, BMTC, and DTC.

7 CONCLUSION

This paper presented a complete predict-then-optimize pipeline for bus frequency allocation, evaluated on a synthetic instance modelled after Pune’s PMPML network. The pipeline integrates a deterministic MIP that respects fleet, depot, and equity constraints with an operator-cost penalty λ tracing the rider-wait / operator-cost Pareto frontier, an XGBoost demand model with quantile heads and split-conformal calibration of the prediction intervals, stochastic and robust formulations driven by quantile-derived scenarios, and a decision-focused neural network trained with an SPO^+ -style surrogate evaluated across 20 random seeds.

Three findings stand out. First, against status-quo frequencies on the synthetic instance the unconstrained optimum ($\lambda = 0$) reduces total waiting time by approximately twenty percent while saturating the fleet; introducing $\lambda > 0$ trades this gain off against bus-period usage along the frontier reported in §4.3. Second, the decision-focused neural network’s regret advantage over the MSE-trained XGBoost is reported as a paired Wilcoxon test across seeds rather than a single-seed difference, and is interpretable both in passenger-hours and as a fraction of the objective. The methodological claim – that prediction quality and decision quality dissociate in predict-then-optimize pipelines – is the contribution; the size of the gap on any particular instance is data-generating-process dependent. Third, the equity floor of three buses per hour at peak for priority routes costs a small fraction of total waiting time, with feasibility under the floor governed by depot-level capacity rather than fleet-level fleet size.

The methodological contribution – how to evaluate predict-then-optimize pipelines without conflating prediction and decision quality, and how to expose the rider/operator trade-off that vanishes when the operator term is omitted – generalises beyond transit. The practical contribution is a turnkey synthetic benchmark that reproduces in approximately 75 minutes; porting it to real operational data on a specific agency

is the natural next step but is not what this paper claims.

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